Compression is one of those computer concepts that quietly shapes almost every digital interaction, yet most users never think about it. From the ZIP files used to bundle and share documents, to the complex WIM format that powers operating system deployment across thousands of enterprise machines, compression technology sits underneath much of how we store and move information.

It started in 1952, when David Huffman invented Huffman Coding — a method that assigned shorter binary codes to frequently occurring characters and longer ones to rare characters. That idea, elegant in its simplicity, became a building block for nearly everything that followed. Abraham Lempel extended the discipline with LZ77 in 1977. Thom Henderson's ARC arrived in 1985. Then, in 1989, Phil Katz released ZIP — and everything changed.

ZIP stood apart from its predecessors for two reasons. First, its compression ratio was meaningfully better. Second, and more consequentially, Katz released it with no license attached. Anyone could implement it, build on it, or distribute it freely. That decision served the world extraordinarily well. It did not serve Katz the same way — a point that deserves its own examination, and one that will be covered in a separate piece.

What followed was an era of building. Some technologies improved on ZIP's open foundation. Others broke from it entirely. As computing evolved, so did the demands placed on compression — and the formats created to meet them.

ZIP

ZIP is both a packaging format and a compression format, designed by Phil Katz and Gary Conway. Because it was released with no license restrictions, it spread everywhere and became compatible with every major platform by default — no additional software required.

The ZIP File Format Specification was first published as part of PKZIP 0.9 in 1989, distributed inside a plain text file called APPNOTE.TXT. That single decision — bundling an open specification directly with the software — is what made ZIP universal. Any developer could read exactly how it worked and implement it themselves.

Under the hood, ZIP uses the DEFLATE compression method, which combines two algorithms: LZ77 and Huffman Coding. LZ77 scans file data for repeated patterns — if the word "compression" appears ten times, subsequent occurrences are replaced with references to the first. Huffman Coding then encodes the result by substituting frequently used characters with shorter bit sequences. That two-stage process is why ZIP performs well on text, source code, and documents, while already-compressed files like MP3s or JPEGs see little benefit.

ZIP's reach extends beyond its own file type. Several widely used formats are, structurally, ZIP files with specific contents inside:

Format	What it contains

.docx, .xlsx, .pptx	Microsoft Office files — ZIP containing XML
.jar, .war	Java application packages
.apk	Android app packages
.epub	Ebook format — ZIP containing HTML
.odt	OpenDocument format — ZIP containing XML
.xpi	Firefox extensions
.nupkg	NuGet packages

This is verifiable directly: rename any *.docx* file to *.zip* and open it. Inside, you will find a structured folder of XML files.

ZIP does have real limitations. Because each file is compressed independently, it finds fewer patterns than formats that treat multiple files as a single continuous stream. The original specification used 32-bit integers for file sizes, capping archives at 4GB — a constraint that ZIP64 later extended, though not all software supports the extension. If the central directory at the end of a ZIP archive is corrupted, recovery is difficult. And the original ZIP encryption scheme is weak enough to be trivially broken; AES encryption requires newer tools that explicitly support it.

None of that undermined ZIP's dominance. It is natively supported on every major operating system, embedded in dozens of file formats, ratified as an ISO standard, and embedded in the foundations of modern software packaging. ZIP did not win by being technically superior. It won by being open, universal, and everywhere — a harder position to compete with than raw performance.

RAR

RAR was developed in 1993 by Eugene Roshal, a Russian software engineer who originally built it for DOS. The name stands for Roshal Archive. Unlike ZIP, Roshal kept the compression algorithm entirely proprietary and never published an open specification. What he did release was decompression source code — with an explicit restriction: it could not be used to reconstruct the compression algorithm. That decision made RAR capable and widely extractable, but never freely creatable. The software is licensed by win.rar GmbH, with copyright held by Alexander Roshal, Eugene's elder brother.

Several capabilities distinguish RAR from its contemporaries:

- **Error Recovery Records** — RAR's most distinctive feature. When creating an archive, RAR can embed recovery data so that if the file is partially corrupted during transfer, it can repair itself. ZIP has no equivalent.
- **Multi-part archives** — RAR was built from early on to split large archives across numbered volumes. This was critical in the era of floppy disks and size-limited email attachments, and one still used today on large download sites.

- **Solid archive mode** — compresses all files together as a single continuous block, finding redundancy across files simultaneously rather than compressing each one in isolation.
- **Digital signing** — RAR supports signing archives so recipients can verify authenticity and detect tampering.
- **Self-extracting archives (SFX)** — RAR can package the archive and a decompressor into a single .exe that the recipient runs without any additional software.

The fundamental strategic weakness is creation. Extracting RAR archives is free, with tools like 7-Zip supporting it without cost. Creating RAR archives requires WinRAR — a paid, licensed product.

WinRAR is technically trial software with a 40-day evaluation period. In practice, the trial never expires; it simply continues reminding users to purchase a license. Millions of people have been running "trial" WinRAR for a decade or more. win.rar GmbH has never enforced expiration, seemingly content with revenue from enterprise customers and the portion of users who choose to pay.

7-Zip

7-Zip was created by Igor Pavlov, a Russian software developer, and first released in 1999. Its native 7z format is open and modular — anyone can examine its specification, and new compression methods can be added without breaking the format. File names are stored as Unicode.

Its compression advantage over competing formats comes from three compounding factors:

- **Massive dictionary** — up to 3.84GB in classic LZMA — allowing it to detect repeated patterns across enormous spans of data.
- **Solid archiving** — compresses all files as a single continuous stream, finding redundancy across files the same way RAR does.
- **Algorithm selection per content type** — bzip2 for certain data, PPMd for text-heavy content, LZMA2 for general use. Rather than forcing one method onto everything, it adapts to what it is compressing.

7-Zip creates and extracts 7z, ZIP, GZIP, BZIP2, XZ, TAR, and WIM archives, and extracts from many others including RAR, ISO, and CAB. One free tool, replacing the need for separate software across nearly every archive format in common use.

7-Zip is available primarily for Windows, with command-line access on Linux and macOS through unofficial ports. It was never locked to one ecosystem by design — only by initial focus.

7-Zip represents what happens when you take ZIP's openness and RAR's compression philosophy and push both further. Bigger dictionaries, better algorithms, completely free, and genuinely open-source — not just "free to extract." It became the format serious users reach for because it has no conditions attached.

CAB

The Cabinet file format was introduced by Microsoft in the mid-1990s, with its first major deployment arriving alongside Windows 95 in 1995. It was originally called Diamond internally. The .CAB extension and the word "cabinet" both derive from Microsoft's internal tooling for the format.

CAB evolved from earlier Microsoft compression techniques used in MS-DOS and early Windows releases to optimize storage on floppy disks. In MS-DOS distributions, dynamic link libraries were frequently compressed into .DL_ files using an LZSS-based algorithm — an early effort to bundle and compress executables that influenced CAB's multi-file archive design.

Where ZIP, RAR, and 7-Zip are general-purpose tools, CAB was purpose-built for one specific job: software installation and distribution on Windows. A CAB file functions as a file library — a single container holding multiple compressed files intended to be copied to a user's system during installation. Compression operates across file boundaries, finding redundancy across files rather than within each file individually, on the same principle as solid archives in RAR and 7-Zip.

Most Windows users encounter CAB constantly without realizing it. Every .msi installer embeds CAB files internally. Every Windows Update package contains them. Almost all Windows hardware drivers ship in CAB format. Windows PE — the stripped-down Windows environment used during OS installation — uses CAB files. .NET Framework installers use them.

WIM

WIM stands for Windows Imaging Format. First introduced in 2007 to support the deployment of Windows Vista, it has since become fundamental to how Microsoft distributes and manages operating systems.

Every format covered before this point is a file archiver. WIM is something different: a disk image format. But where formats like ISO or VHD copy a disk sector by sector — duplicating every byte regardless of whether it contains data — WIM operates at a higher level. It packages files as an intelligent archive rather than photographing a disk. That distinction is why a Windows installation image is a manageable 4–6GB rather than a raw sector copy that would be far larger.

WIM's most significant capability is single-instance storage. When multiple files in an image are identical, WIM stores that file exactly once and references it from every location that needs it. The more images share common files, the more pronounced the space savings become.

This is why a single install.wim file can contain multiple editions of Windows — Home, Pro, Education, Enterprise — without growing proportionally in size. All four editions share thousands of identical system files. WIM stores each shared file once. The result is that four editions of Windows cost far less than four times the space of one.

A single WIM file can contain multiple discrete images, each referenced by a numerical index or a unique name. This is why when you boot a Windows installer and choose between editions, all those options are already present inside one file.

The operational value extends to enterprise environments. WIM files can be mounted, modified, and unmounted without applying or rebuilding the entire image. An IT department managing thousands of computers can mount a WIM, inject updated drivers and security patches, unmount it, and deploy the result across every machine — without a live installation on any of them.

WIM is the most technically sophisticated format in this series. Every other format compresses files. WIM compresses an entire operating system, stores multiple editions simultaneously without redundancy, signs the package cryptographically, allows modification without reinstallation, and enables one file to be deployed identically across thousands of different hardware configurations. It is the reason a 4GB download installs into 20GB, the reason Windows Home and Pro ship in the same ISO, and the reason enterprise IT can manage fleets of machines from a single image file.

Comparison

Origin & Purpose

	ZIP	RAR	7-Zip	CAB	WIM
Created by	Phil Katz & Gary Conway	Eugene Roshal	Igor Pavlov	Microsoft	Microsoft
Primary use	Universal file sharing	Download distribution	Maximum compression storage	Software installation packaging	OS deployment & imaging

Compression

	ZIP	RAR	7-Zip	CAB	WIM
Compression ratio	Weakest	Good	Best	Moderate	Very good
Solid archive	No	Yes	Yes	Yes (cross-file)	Yes

Compress speed	Fastest	Moderate	Slowest	Fast	Moderate
Decompress speed	Very fast	Fast	Fast	Fast (LZX optimised)	Fast

Features

	ZIP	RAR	7-Zip	CAB	WIM
Error recovery	No	Yes	No	No	No
Single-instance storage	No	No	No	No	Yes
Random access	Yes	No (solid)	No (solid)	Yes	Partial

Access & Cost

	ZIP	RAR	7-Zip	CAB	WIM
Create archives	Free	Paid (WinRAR)	Free	Free (Windows Tools)	Free (DISM / ImageX)
Extract archives	Free everywhere	Free everywhere	Free everywhere	Free (Windows)	Free (Windows)

End of Part 1. Part 2 coming soon.